

BAYESIAN SURVIVAL, LONGITUDINAL AND JOINT MODELS WITH INLA - JOINT MODELS

Denis Rustand - Research Scientist

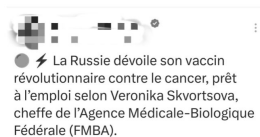
Bordeaux Population Health Research Center
National Institute of Health and Medical Research (Inserm UMR 1219), France

València International Bayesian Analysis Summer School, 9th edition

Contact: denis.rustand@u-bordeaux.fr

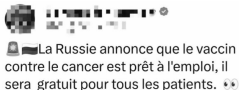
SINCE TUMORS ARE A DIRECT MANIFESTATION OF
CANCER, A GOOD TREATMENT SHOULD SIMPLY
REDUCE TUMOR SIZE?

A CASE STUDY IN TUMOR SHRINKAGE



1:55 · 07 sept. 25 · 193K Vues

659 reposts 40 citations 2 277 J'aime



13:38 · 07 sept. 25 · 360K Vues

720 reposts 49 citations 3 958 J'aime



TOI Lifestyle Desk / etimes.in / Updated: Sep 11, 2025, 08:12 IST



Russia unveils a REVOLUTIONARY cancer vaccine now ready for clinical use — FREE for patients.

FMBA chief Skvortsova: "3 years of trials PROVED safety & high efficacy."

10:17 · 07 sept. 25 · 1,3M Vues

8 360 reposts 371 citations 42,7K J'aime



Russia's Cancer Vaccine Enteromix Displays 100 Per Cent Efficacy In Clinical Trials

Curated By : Manisha Roy | Sep 07, 2025, 00:02

Enteromix, a personalized mRNA cancer vaccine, showed 100% efficacy and safety in trials. Developed by Russian institutes, it awaits Ministry of Health approval for public use.



"Reduction in tumor size ranging from 60% to 80%."

Veronika Skvortsova, Russian state agency (Sept 2025)

THE LIMITATIONS OF TUMOR SIZE AS AN ENDPOINT

The “cancer cure” from Russia: When data is misleading.

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No Safety Data: No Phase I trials.

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JOURNAL ARTICLE

Estimation of clinical trial success rates and related parameters

Chi Heem Wong, Kien Wei Siah, Andrew W Lo 

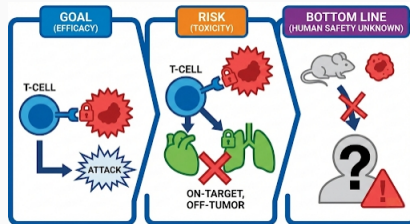
Biostatistics, Volume 20, Issue 2, April 2019, Pages 273–286,

<https://doi.org/10.1093/biostatistics/kxx069>

Published: 31 January 2018 **Article history** ▼

*Even after successful animal testing, **96.6%** of oncology drugs fail in human trials.*

THE BLIND SPOT OF TUMOR SHRINKAGE



- Reactivated T-cells identify and **attack cancer cells** based on a **specific antigen marker**.
- The same T-cells **attack healthy organs** (Heart, Lungs) because they share the **same antigen marker**.

If we only look at the tumor size, the drug looks efficient.

If we only look at survival, the drug looks like a failure.

→ What about a method that sees **both simultaneously**?

Unifying the Data: Simultaneously estimates the Longitudinal process (SLD) and the Survival process.

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Dynamic Risk Assessment: Treats tumor size as an evolving predictor (as the tumor grows, the hazard of death or progression updates in real-time).

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Correcting Bias:

- **Informative Censoring:** Corrects for patients dropping out due to toxicity or rapid progression (where SLD data stops non-randomly).
- **Measurement Error:** Uses the estimated true trajectory rather than noisy raw measurements.

TREATING HETEROGENEITY AS NOISE

Randomized trials are unbiased, but can be inefficient.

▼ Standard Analysis (Cox/RECIST)

Summarizes, then Analyzes

- › Collapses complex SLD history into a single "+20%" event.
- › Treats individual heterogeneity as noise to be averaged out.

↪ Information Loss = Lower Power

- More patients exposed to placebo/control arms.
- Effective drugs take longer to reach the market.

≡ Joint Modeling

Analyzes the Full History

- › Models the continuous SLD values over time.
 - › Reduces unexplained variance, increase the signal-to-noise ratio.
- ↪ Explains heterogeneity to gain power
- Transforms routine tumor measurements into statistical gain.

MODELS OF INTEREST

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LONGITUDINAL SUBMODEL

$$\text{SLD} = \underbrace{\text{Fixed Effects}}_{\substack{\text{Average}(t) \\ \text{(Includes treatment effect!)}}} + \underbrace{\text{Random Effects}}_{\text{Indiv. Deviation}(t)} + \underbrace{\epsilon}_{\text{Error}}$$

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(Includes treatment effect!)

SURVIVAL SUBMODEL

Standard Cox

(Treats heterogeneity as statistical noise. Adjusts only for baseline covariates)

$$h_i(t) = h_0(t) \cdot \exp(\beta \cdot \text{Trt} + \gamma \cdot \mathbf{X}_{\text{Baseline}})$$

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Joint model with Shared

Random Effects (SRE)

(Adjusts for individual deviation from the population average)

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Joint model with

Current Value (CV)

(Links risk to longitudinal tumor size)

$$h_i(t) = h_0(t) \cdot \exp(\dots + \alpha \cdot (\text{Average}(t) + \text{Deviation}_i(t)))$$

Hazard ratio of treatment

Standard Cox

$$HR = \exp(\beta)$$

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Joint Model with Shared Random Effects (JM SRE)

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Joint Model with Shared Random Effects (JM SRE)

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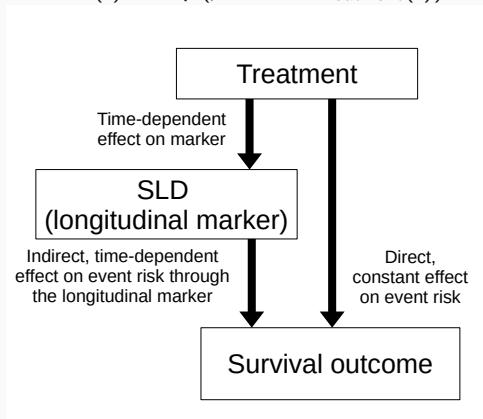
Joint Model with Current Value (JM CV)

$$HR(t) = \exp(\beta + \alpha \cdot \Delta_{\text{Treatment}}(t))$$

↳ Time-dependent!

Joint Model with Current Value (JM CV)

$$HR(t) = \exp(\beta + \alpha \cdot \Delta_{\text{Treatment}}(t))$$



STATE OF THE ART METHODOLOGY

The issue with "Constant treatment effect"

STATISTICS IN ONCOLOGY | May 03, 2010



Basic Concepts and Methods for Joint Models of Longitudinal and Survival Data

Authors: Joseph G. Ibrahim, Haipeng Chen, and Li-Shu M. Chen

[AUTHOR INFO & APPLICATIONS](#)

J Clin Oncol 28, 2799-2801 (2010) • Volume 28, Number 18 • DOI: 10.1200/JCO.2009.25.0954

trajectory function of the model. The trajectory function is typically specified as a linear function of time t_{ij} and a treatment indicator Z_p given by $X_{ij} = \theta_{0i} + \theta_{1i} \times t_{ij} + \gamma Z_p$ or as a quadratic function of time.

The Reality of RCTs

- 🕒 **At Baseline (Time = 0):**
Groups are identical due to randomization.
- 🕒 **Over Time (Time > 0):**
The drug takes effect. The curves diverge.

Ibrahim et al. (time + treatment)



Our approach (time x treatment)



WHY NOT JUST PUT SLD IN A COX MODEL?

ENDOGENOUS MARKERS

Air pollution is an **external (exogenous)** time-dependent covariate. Tumor size is different: it is part of the patient's disease process **(internal/endogenous)**.

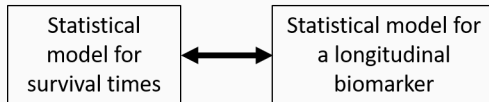
- treatment changes SLD
- SLD may change death risk
- death prevents future SLD measurements
- SLD is measured with error and at irregular visits

The observed SLD history is not an external exposure history.

JOINT MODELING LONGITUDINAL AND SURVIVAL DATA

TIME-DEPENDENT VARIABLES IN SURVIVAL

The proportional hazards model can handle time-dependent covariates but the likelihood requires **knowing the value of these covariates for all subjects at risk for each event time**.



A joint model uses a regression model to get the value of the longitudinal marker at any time point and can **take advantage** of the **complementary information** flowing between the longitudinal and survival outcomes.

Joseph G. Ibrahim et al. Basic Concepts and Methods for Joint Models of Longitudinal and Survival Data. Journal of Clinical Oncology 2010.

JOINT MODEL OF LONGITUDINAL AND SURVIVAL DATA

- Models disease as a **complex, dynamic system**.
- Links **repeated biomarker measurements** with **time-to-event** data.
- Captures variables whose **trajectories correlate with event status**.
- Addresses bias from subjects **dropping out** due to events.
- Accounts for **noise** in longitudinal data.
- Enhances power by **using all data sources** simultaneously.
- Outperforms simplified or two-stage approaches.

JOINT MODELING - EXAMPLES

A few examples of recent applications of joint models:

- Cancer tumor dynamics and the risk of death
- Analysis of CD4 lymphocytes counts and AIDS survival
- Prostate-specific antigen dynamics and the risk of cancer recurrence
- Dynamics of aortic gradient and aortic regurgitations and their relationship with the competing risks of death or reoperation in the field of cardiac surgery
- Cognitive markers's relationship with the time to onset of Alzheimer's disease
- Jointly modeling forest fires ignition, number of fires and the proportion of burned area in Portugal

Random effects used to fit the longitudinal marker need to be **integrated out** in the likelihood.

A **second level of numerical integration** is required when the risk function includes time-dependent variables.

$$\text{Survival likelihood} = \prod_{i=1}^N S_i(t) \lambda_i(t)^{[\text{Event indicator}]}$$

where $S(t) = \exp(-\int_0^t \lambda(u) du)$ and $\lambda_i(t) = \lambda_0(t) \exp(X_i(t) \gamma)$

An Overview of Joint Modeling of Time-to-Event and Longitudinal Outcomes. Grigorios Papageorgiou, Katya Mauff, Anirudh Tomer, and Dimitris Rizopoulos. Annual Review of Statistics and Its Application 2019

JOINT MODEL WITH INLA

- Modular design that uses univariate models as **flexible building blocks**.
- Computation **speed**
- **Robust** estimation for highly parameterized models.
- **INLAjoint** simplifies thousands of lines of code into **concise syntax**.

THE COMPUTATIONAL BOTTLENECK

Why are joint models historically hard to fit?

- The likelihood requires **integrating out** the random effects.
- As models grow (multivariate markers, splines), this becomes a **high-dimensional integral**.

Traditional Approaches (MCMC / Quadrature):

- Can take **days** to run.
- Prone to **convergence failures**.
- Forces researchers to choose **overly simple models**.

INLA:

- Uses **analytical approximations** for latent Gaussian models.
- Bypasses sampling entirely: turns days of computation into **minutes**.

TWO SUBMODELS, ONE LIKELIHOOD

A standard joint model for a **longitudinal biomarker** and **survival times** is defined as follows:

$$\begin{cases} Y_{ij} = X_{ij}^{\top} \boldsymbol{\beta} + Z_{ij}^{\top} \mathbf{b}_i + \epsilon_{ij} & \text{(Biomarker)} \\ \lambda_i(t) = \lambda_0(t) \exp(X_i(t)^{\top} \boldsymbol{\gamma} + h(\cdot)) & \text{(Time-to-event)} \end{cases}$$

The association between the longitudinal and survival part can take various forms.

CONDITIONAL INDEPENDENCE

How do we actually combine these two distinct processes?

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^N \int \underbrace{p(\mathbf{y}_i \mid \mathbf{b}_i, \boldsymbol{\theta})}_{\text{Longitudinal}} \times \underbrace{p(T_i, \delta_i \mid \mathbf{b}_i, \boldsymbol{\theta})}_{\text{Survival}} \times \underbrace{p(\mathbf{b}_i \mid \boldsymbol{\theta})}_{\text{Random Effects}} d\mathbf{b}_i$$

- Given the random effects, the longitudinal measurements and the survival time are **independent**.
- The random effects act as the **bridge** that forces the two submodels to share information.

ASSOCIATION STRUCTURE: SHARED RANDOM EFFECTS

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha_0 b_{i0} + \alpha_1 b_{i1}\} \end{cases}$$

- Links risk to **patient-specific** latent traits (**independent** intercept and slope).
- **Computationally efficient**: avoids the complexity of time-dependent components while capturing heterogeneity.
- Asks if a patient's **baseline** marker value or **growth rate** is linked to survival outcomes.

ASSOCIATION STRUCTURE: INDIVIDUAL DEVIATION

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha(b_{i0} + b_{i1}t)\} \end{cases}$$

- Links risk to the **evolving individual deviation** from the population average at time t .
- Combines baseline and slope into **a single, time-varying deviation**.
- Isolates **residual heterogeneity**, without the effect of observed covariates like treatment.

ASSOCIATION STRUCTURE: CURRENT VALUE

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_{it} \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha \eta_i(t)\} \end{cases}$$

- Links risk to the **"true" underlying** biomarker level at time t .
- **Reduces bias**: filters out measurement error and noise from the observed data.
- Useful when **the marker is the biological driver**

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha \frac{d\eta_i(t)}{dt}\} \end{cases}$$

- Links risk to the current **rate of change** of the biomarker.
- Distinguishes a large stable tumor from a smaller but rapidly growing one.
- Asks if the speed of **disease progression** is an independent, prognostic factor of survival.

ASSOCIATION STRUCTURE: CUMULATIVE VALUE (AUC)

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp \left\{ \gamma trt_i + \alpha \int_0^t \eta_i(s) ds \right\} \end{cases}$$

- Links risk to the **cumulative exposure** (Area Under the Curve) of the biomarker up to time t .
- Accounts for the **accumulated burden over time** rather than concomitant association.
- Asks if exposure (or **lagged exposure** within specific past windows) drives the event risk.

ASSOCIATION STRUCTURE: NON-LINEAR ASSOCIATION

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp \{ \gamma trt_i + f(\eta_i(t)) \cdot \eta_i(t) \} \end{cases}$$

- Links risk using a **flexible, data-driven function** (splines/random walks) rather than a restrictive linear assumption.
- Captures U-shaped, inverted-U, or threshold risk profiles **where linear models fail**.
- Asks if extreme values in both directions have **distinct** prognostic weight.

DYNAMIC PREDICTION

A standard Cox model gives a **static** survival curve at baseline.

A Joint Model enables **dynamic, real-time** risk updating.

- **Month 0:** Patient starts therapy. We predict their baseline survival curve.
- **Month 3:** Patient comes for a scan. Tumor has shrunk. We **update** the survival curve based on this new data.
- **Month 6:** Patient comes for another scan. Tumor is growing. The survival curve **dynamically drops**.

→ Joint models pave the way for **personalized, real-time medicine**.

THE DATA ALIGNMENT CHALLENGE

Joint models must combine two different data structures:

- **Longitudinal Data:** Many rows per patient. Measured discretely at specific visits (e.g., months 0, 3, 6, 9).
- **Survival Data:** One row per patient. Measured continuously (e.g., event at month 7.2).

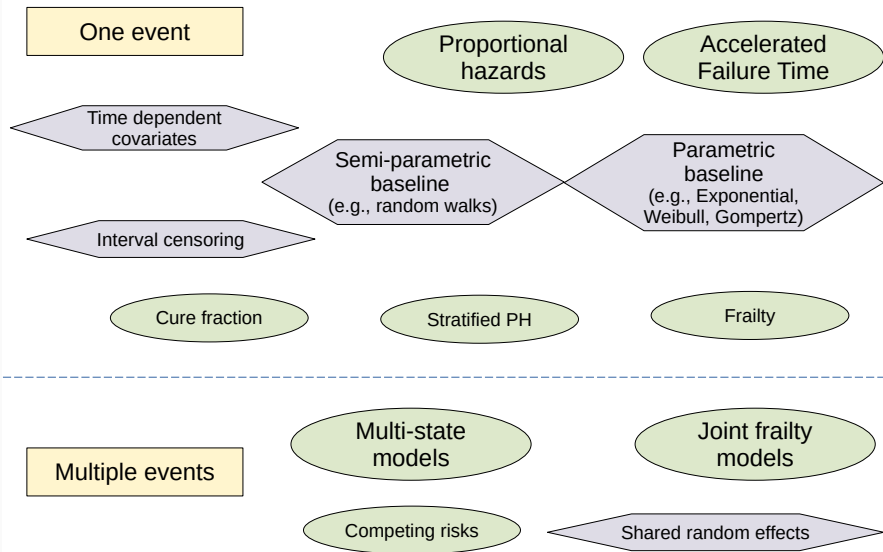
The INLAjoint approach:

- Automatically handles the data expansion.
- Discretizes the follow-up time into small intervals.
- Aligns the longitudinal expected value to the exact interval where the survival risk is evaluated.

BACK TO THE CODE

SUMMARY OF SURVIVAL AND LONGITUDINAL
MODELS AVAILABLE WITHIN THE INLA/INLAJOINT
FRAMEWORK

SURVIVAL MODELS FOR TIME-TO-EVENT DATA



LONGITUDINAL MODELS FOR REPEATED MEASUREMENTS

